

Variation in Many-Analyst Projects: Problematic or Helpful? A Multiverse Approach to Address Different Sources of Variation

Thesis Proposal

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Methodology and Statistics for the Behavioural, Biomedical and Social Sciences

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Journal?

word count: 742

October 10, 2025

1 Introduction

Recent studies have shown considerable variation in outcomes when multiple researchers answer the same question using the same data, raising concerns about the reproducibility of science. For example, Kowall et al. (2025) asked 16 teams to estimate the effect of marital status on cardiovascular outcomes, and the results varied widely from negative to positive effects (e.g., odds-ratios ranged from 0.72 — 1.29).

Many-analyst projects can be useful to gain insight into both natural variation in science, and analysis robustness. However, various studies have failed to identify where this variation stems from. According to Silberzahn et al. (2018), analysts' prior beliefs, level of experience or peer-rated quality of the analysis do not seem to explain the variation. Additionally, Hoogeveen et al. (2023b) found that most researchers used similar statistical models, but their variable definitions varied widely. On the other hand, Kowall et al. (2025) stated that variation likely stemmed from the choice of longitudinal versus cross-sectional data, rather than variable definitions or statistical models. Interpreting variation within and across many-analyst projects is difficult – is there a "hidden universe" of uncertainty that poses a problem for credible and reproducible science or is the variation meaningful and can it drive the advancement of theories?

Table 1: Sources of variation according to Auspurg and Brüderl (2024)

Problematic	Helpful
(a) Model uncertainty, including subjective choices in data handling, and choosing a type of statistical model	(c) Substantively meaningful variation, based on different definitions of dependent, independent and co-variables and/or target populations
(b) Coding Errors	(changes over time)

According to Auspurg and Brüderl (2024), three different sources of variation can be distinguished in many-analyst projects. As shown in table 1, Auspurg and Brüderl (2024) classify (a) and (b) as 'problematic', because they can introduce uncertainty in scientific analyses, making conclusions less credible. Source (c), on the other hand, can be seen as 'helpful', because it can provide information about substantive variation that actually exists in the population. According to Simonsohn et al. (2015), "regressions with and without a certain set of covariates (...) are different answers to different questions."

Research Question

How do different sources of variation, including handling missing data, choice of statistical model and the definition of outcome variables, covariates and target population (longitudinal versus cross-sectional), contribute to variation in many-analyst projects?

Expectations

Similar to previous multi-analyst studies, I expect considerable variation depending on analytical choices. Because the explanations for the variation in previous studies varied widely, and may differ across studies, I have reservations about making explicit expectations in this regard. However, since I will be basing my study on Kowall et al. (2025), I expect to potentially find similar patterns (where the type of data seemed to have most impact on the estimates) — although I assume this with great caution, because they only included 16 teams and my multiverse analysis (explained in the next section) will be more extensive.

2 Analytic Strategy

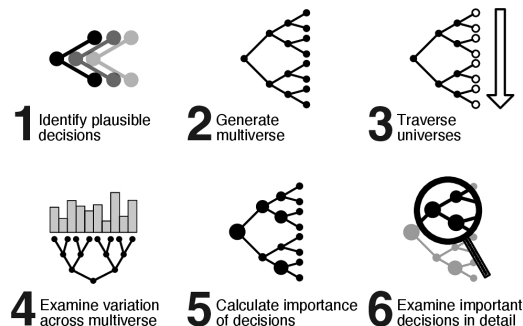


Figure 1: Comprehensive Overview of Multiverse Analysis, from Simson et al. (2024)

Although many-analyst projects allow quantifying variance resulting from analytic decisions, this is not done systematically. Multiverse analysis is a way to systematically map a wide variety of reasonable ‘research paths’, aiming to increase transparency about subjective choices and their impact on the results (Steege et al., 2016). This method allows distinguishing different sources of variation and evaluating their impact, which is the goal of this thesis.

Because the study by Kowall et al. (2025) includes many different sources of variation, I decided to base my multiverse analysis on this study. The choices made by analysts, that I will use, can be found in table 2. Figure 1 shows an overview of the steps I will follow in my multiverse (Simson et al., 2024).

Table 2: Different choices that were made by analysts in the study by Kowall et al. (2025). The multiverse analysis in this study will be based on these choices.

(a) Potentially problematic sources of variation

Missing Data Handling Methods	Type of Model
Complete cases (16)	Cox Proportional Hazards model (6)
Imputed data from SHARE (3)	Logistic Regression (5)
"No information" (1)	Log-binomial model (2) Mixed-effects logistic regression (1) Discrete time mixed effect model (1) Generalized estimating equations (1)

(c) Potentially meaningful sources of variation

Outcome (Dependent Variable)	Adjustment Set (Covariates)	Type of Data (change over time)
Heart attack / stroke ever (16)	Age (12)	Strictly longitudinal (10)
Age at heart attack / stroke (9)	Sex (12)	Longitudinal with wave 1 as the only baseline (3)
Heart attack / stroke since last interview (7)	Country (7)	Strictly or partly cross-sectional (6)
Main cause of death heart attack / stroke (5)	Smoking (6) Education (4) Physical activity (4)	

Note: The numbers in brackets behind the decisions denote how many analysts made this choice in their analysis.

Note: The adjustment sets in the original study by Kowall et al. (2025) contained 25 variables. When including all, the multiverse analysis would become too complex, as there would be over 33 million possible combinations. Therefore, I decided to only include covariates that were used by at least four (or more) analysts in the original study, except the 'no covariates' category which will also be included. I am aware this is an arbitrary decision.

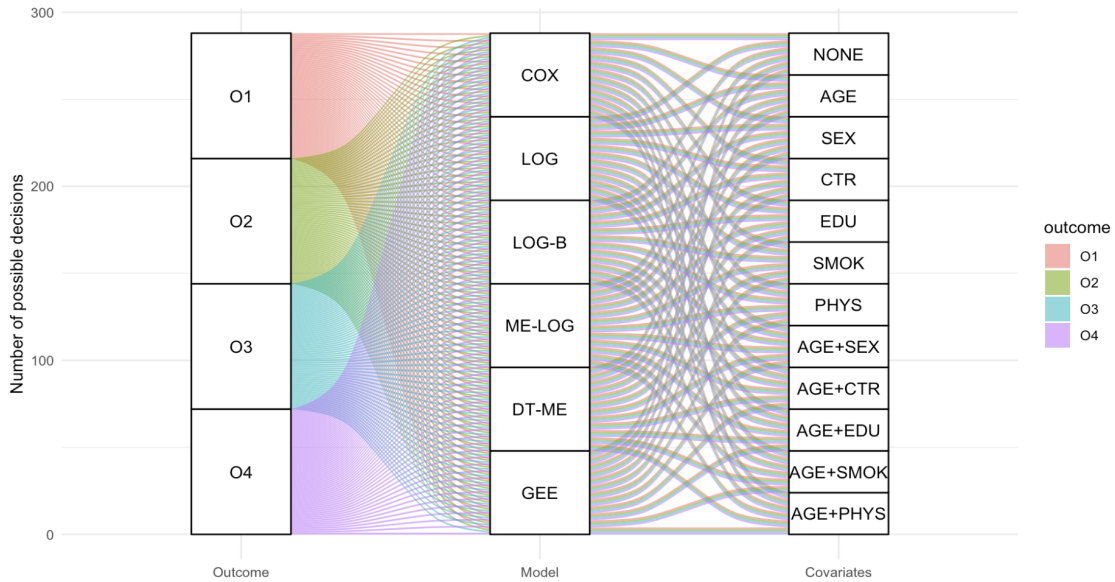


Figure 2: Oversimplified overview of the multiverse analysis in this study. This figure shows only a part of the multiverse; many more combinations of both the outcome variables and the covariates are possible. *O1-O4: outcome variables, as described in table 2; COX: Cox Proportional Hazards model, LOG: logistic regression, LOG-B: log-binomial model, ME-LOG: mixed-effects logistic regression, DT-ME: discrete time mixed effect model, GEE: generalized estimating equations; CTR: country, EDU: education, SMOK: smoking, PHYS: physical activity.*

Figure 2 shows that — even when looking at a small part of the multiverse — there are many different paths to estimate the effect of marriage on cardiovascular disease. When looking at table 3, it becomes clear that there are large numbers of possible analyses. Since this set of reasonable paths is too large to estimate all effect sizes within the timeframe of this project, based on Simonsohn et al. (2015), I decided to randomly sample a subset of **N** possible specifications.

For assessing the variance in the estimates obtained with the multiverse analysis, I will look at the distributions of the estimates, and compare this to the distribution of the estimates in the original many-analyst project.

Table 3: Number of possible analysis paths based on various decision patterns.

Decision	Unique possibilities	Number of combinations
Missingness	2	2
Model Type	6	6
Outcome	4	$2^4 = 16$
Adjustment Set	7	$2^7 = 128$
Data Type	3	3
Total		73.728
If missingness is disregarded; and only two data types are used (strictly longitudinal/cross-sectional).		
		24.576
If missingness is disregarded; only two data types; and only covariates that at least 5 analysts chose are used.		
		12.288
If missingness is disregarded; only two data types; only covariates that at least 5 analysts chose; and only outcome variables that at least 7 analysts chose are used.		
		6.144
If missingness is not disregarded; only two data types; only covariates that at least 5 analysts chose; only outcome variables that at least 7 analysts chose; and only the models that at least 2 analysts chose are used.		
		3.072
If missingness is disregarded; only two data types; only covariates that at least 5 analysts chose; only outcome variables that at least 7 analysts chose; and only the models that at least 2 analysts chose are used.		
		1.536

Data

Like Kowall et al. (2025), I will use the Survey of Health, Ageing and Retirement in Europe (SHARE) dataset: a large, freely available (after registration), panel study conducted in 28 European countries in different waves (in the original study 1-7 were included).

Ethics

This project is under review by the Faculty Ethics Review Committee (FETC) of the Faculty of Social Sciences, with registration number: XXX.

Software

Table 4: All analyses will be performed in RStudio (Team). I will likely use (at least) the following:

Package	Citation
tidyverse	(Wickham et al., 2019)
ggalluvial	(Brunson, 2020)
multiverse	(Sarma et al., 2021)
specr	(Masur and Scharkow, 2020)
lme4	(Bates et al., 2015)
survival	(Therneau, 2024)
geepack	(Halekoh et al., 2006)

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